

Abid

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FULL STACK AI/ML Lead

Summary:

Lead AI/ML Developer with 9 years of experience in AI-powered applications, scalable APIs (FastAPI, Django, Flask), and ML model integration using TensorFlow, PyTorch, and Scikit-learn. Expertise in AI lifecycle governance, AI/ML/GenAI policies, MRM, MRO, RCSA, and NIST AI Risk Management Framework. Skilled in cloud computing (AWS, GCP), containerization (Docker, Kubernetes), and CI/CD automation (GitHub Actions, Terraform). Passionate about AI risk management, secure deployments, and scalable AI-driven solutions.

Work History

Lead AI/ML Developer

July 2021 - Present

Delta Air Lines

- Spearheaded the development and deployment of AI-powered applications, integrating machine learning models into scalable web platforms using FastAPI, Flask, and Django for backend and React for frontend.
- Architected and optimized APIs for AI model serving, ensuring seamless data flow and real-time inference performance.
- Built and maintained end-to-end ML pipelines, leveraging TensorFlow, PyTorch, and Scikit-learn to enhance model training, evaluation, and deployment efficiency.
- Designed AI lifecycle governance frameworks, ensuring compliance with AI, ML, GenAI policies, standards, and procedures while mitigating risks through Model Risk Management (MRM) and Model Risk Oversight (MRO).
- Developed and implemented AI risk controls, integrating NIST AI Risk Management Framework (RMF) and conducting RCSA (Risk Control Self-Assessment) for AI capabilities.
- Engineered ETL pipelines for processing structured and unstructured data, enhancing AI model performance and decision-making capabilities.
- Led cloud-based AI model deployments, utilizing AWS services (S3, Lambda, SageMaker) to ensure high availability and scalable AI solutions.
- Containerized AI applications with Docker and Kubernetes, automating deployments and enabling seamless scaling across cloud environments.
- Automated CI/CD pipelines with GitHub Actions, streamlining AI model releases and reducing deployment cycles.
- Enhanced AI model interpretability and fairness, applying SHAP, LIME, and AUC-ROC techniques for risk monitoring and bias detection.
- Integrated big data solutions with Snowflake, BigQuery, and vector databases, enabling large-scale AI applications with real-time processing.
- Provided technical leadership and governance, collaborating with compliance, security, and risk management teams to drive responsible AI adoption while mentoring engineers on AI best practices.

Python Developer (AI&APIs)

April 2018 - June 2021

Fidelity Investments

- Designed and optimized high-performance RESTful APIs using FastAPI, Flask, and Django, ensuring low-latency and scalable service communication.
- Led the development of microservices architecture, improving modularity and maintainability of backend systems.
- Built efficient data processing pipelines for structured/unstructured data, integrating vector databases and optimizing ML model inputs.
- Designed and managed data workflows using PostgreSQL and MongoDB, improving query performance and data integrity.

- Developed and automated CI/CD pipelines using GitHub Actions, Docker, and Kubernetes, accelerating deployment cycles.
- Implemented API security best practices, including OAuth2, JWT authentication, and rate limiting to enhance system security.
- Integrated machine learning models into production APIs, optimizing inference speed and resource efficiency.
- Worked with AWS services (Lambda, S3, EC2) for cloud deployment, ensuring scalability and cost optimization.
- Collaborated with ML engineers and data scientists to deploy and monitor models in production, ensuring seamless API integration.
- Spearheaded infrastructure improvements, leveraging containerization (Docker, Kubernetes) and orchestration tools to enhance system reliability.
- Mentored junior developers, conducted code reviews, and contributed to architectural decisions to drive technical excellence.
- Managed tasks using Agile methodologies, utilizing JIRA and Confluence for streamlined project execution.

Backend Developer
JPMorgan Chase & Co

October 2016 - March 2018

- Built and optimized scalable, RESTful APIs using Django and Flask, ensuring seamless data flow and system efficiency.
- Designed and maintained PostgreSQL databases, optimizing queries for better performance and reliability.
- Integrated third-party services like Stripe, Mapbox, and HubSpot, streamlining payment processing and location-based functionalities.
- Developed and maintained background task processing using Celery, improving application responsiveness and task execution.
- Implemented secure authentication and authorization mechanisms, strengthening system security and user data protection.
- Automated CI/CD pipelines with GitHub Actions and Azure DevOps, reducing deployment friction and enhancing development velocity.
- Containerized applications with Docker and orchestrated services using Kubernetes, improving scalability and deployment efficiency.
- Worked closely with frontend teams using React/Angular, ensuring smooth integration and API usability.
- Followed test-driven development (TDD) practices to maintain high code quality.
- Gained hands-on experience with AWS services like S3, Lambda, and RDS for cloud-based applications.
- Worked within Agile methodologies, managing tasks and progress using JIRA and Confluence.
- Contributed to architectural decisions and best coding practices to enhance system scalability.

Education
Bachelor's in computer science

Skills

Python	SQL	Scrapy	TypeScript	Django	FastAPI	SQLAlchemy	OpenAI API	NLTK
Fine-Tuning	ROC	MongoDB	NoSQL	AWS	end-to-end AI/ML	cloud computing	JIRA	
Leadership	REST & GraphQL APIs	Model Versioning	OOP	GitHub Actions	Grafana	team work		
Stakeholder Management	Adaptability	Code Quality	Project Management	ETL Pipelines				

- Certification:**
- **Certified Kubernetes Administrator (CKA)**
 - **Microsoft Certified: Azure AI Engineer Associate**
 - **Google Cloud Professional Machine Learning Engineer**

Projects:

- **Project 1: AI-Powered Code Review System**

Developed a machine learning-based code review system using Python, leveraging NLP techniques for automated code analysis and suggestions. Integrated with GitHub APIs to analyze code repositories and pull requests, improving developer productivity by providing instant feedback.

- **Project 2: Security Anomaly Detection using Machine Learning**

Designed a machine learning model to detect anomalies in security logs. Utilized Python, Scikit-learn, and XGBoost to create a model that automatically identifies unusual patterns in large datasets, enhancing the security of client platforms.

- **Project 3: Image Recognition for Media Content Management**

Developed a deep learning-based image recognition system to classify and categorize images in a media content management platform using CNNs (Convolutional Neural Networks). Automated the media tagging process, improving content discovery and management.